



Indoor Air Quality in Critical Indoor Environments: A Review Paper

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Abstract Indoor air quality (IAQ) plays a critical role in public health, particularly in environments where individuals spend prolonged periods of time indoors, and this become a key consideration to researchers and health practitioners. This comprehensive review investigates and highlights recent studies, conducted in diverse geographic regions, and published between 2019 and 2024 in peer-reviewed journals on indoor air quality (IAQ) in critical indoor environments such as healthcare facilities, schools, and residential buildings. The purpose of this review is to draw out recent key findings and highlights the gaps associated in air quality studies through a methodological analysis of a series of research articles. Major pollutants identified in such review study include particulate matter, carbon dioxide, carbon monoxide, volatile organic compounds, formaldehyde, and biological contaminants. This study highlights how poor ventilation, seasonal variations, building design, and daily indoor activities contribute to elevated pollutant levels. Evidence shows that these exposures are linked to a range of health

outcomes, including respiratory illnesses, cognitive decline, and increased risks for vulnerable populations such as children and patients. The review highlights factors affecting IAQ resulting in exacerbated health risks, in addition to different air quality management strategies providing control programs and special requirements especially for critical indoor settings, where maintaining specific environmental conditions is essential to ensure safety, health, comfort, or proper functioning of sensitive processes or equipment. Considering future developments in IAQ, this review paper provides a forward-looking perspective and a comprehensive air quality management framework to serves as a valuable resource for researchers, engineers, practitioners, and policymakers.

Keywords Indoor air quality · Residential buildings · Hospitals · Schools · Chemical air pollutants · Microbiological contaminants · Air quality indicators

1 Introduction

Indoor air pollution (IAP) is a global concern that requires acknowledgment and action as it exacerbates health conditions of indoor occupants, in general, and vulnerable individuals like children and elderly people in particular (Morawska et al., 2020). Indoor air pollution is still a leading cause of premature and preventable death worldwide (Cunha-Lopes et al., 2019).

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According to World Health Organization (WHO), poor IAQ is accountable for 3.8 million deaths worldwide, annually (Abulude et al., 2023). IAQ is associated with 32% of heart disease cases, 23% of stroke cases, 21% of lower respiratory infection cases, 6% of lung cancer cases and 19% of cases of chronic obstructive pulmonary disease (COPD) are recognized as the most prevalent conditions associated with indoor air pollution (Abulude et al., 2023; Chuanqi & Qianqian, 2020; World Health Organization (WHO), 2024). Indoor air quality (IAQ) not only affects the physical health of indoor occupants (Bueno de Mesquita et al., 2022; Fazli et al., 2021; Kelly & Fussell, 2019; Vardoulakis et al., 2020), but also disturbs various behavioural characteristics, sleeping habits, learning skills, and cognitive abilities (Morawska et al., 2020; Tran et al., 2020; Zhang et al., 2020). In the past few years, due to COVID-19 pandemic and online business market, research work on air quality control has shifted focus into indoor spaces, identifying these behavioural and lifestyle changes with increasing levels of people staying indoors (Coelho et al., 2021; Fazli et al., 2021). Common preventable risk factors for poor IAQ include pollutants from indoor sources that are mainly attributed to anthropogenic activities such as smoking, electronic cigarettes (ECs), cooking, synthetic paints, and emissions from fomites including furniture and construction materials (Bueno de Mesquita et al., 2022). Harmful biological pollutants include bacteria, viruses, and fungi, while chemical pollutants comprise of formaldehyde (CH_2O), benzene (C_6H_6), ozone (O_3), nitrogen oxide (NO), nitrogen dioxide (NO_2), particulate matter (PM_{10} and $\text{PM}_{2.5}$), carbon monoxide (CO), carbon dioxide (CO_2) and total volatile organic compounds (TVOCs) (Fazli et al., 2021).

Critical indoor environments are specific closed environments where occupants and vulnerable individuals, including elderly, children, pregnant women, and individuals with chronic disease are at risk of exposure to indoor pollutants, chemical and biological contaminants. Such settings include health-care facilities, schools, and residential homes where people spend at least 8 h a day and are frequently exposed to indoor air contaminants (Coelho et al., 2021; Fazli et al., 2021; Tran et al., 2020; Zhang et al., 2020). Therefore, indoor environments should safeguard and enhance occupant's health, comfort, and productivity, as people spend around 90% of

their lives indoors (Coelho et al., 2021; Fazli et al., 2021; Kelly & Fussell, 2019; Tran et al., 2020; Zhang et al., 2020). However, there is still limited knowledge regarding the causes of symptoms observed in nonindustrial indoor settings such as office buildings, recreational facilities, schools, hospitals and residencies where indoor environmental quality (IEQ) becomes a major factor that determines the comfort of indoor occupants. Indoor environmental quality (IEQ) is influenced by multifactor parameters such as air contaminants, thermal comfort, and psychosocial issues. There is an increasing amount of scientific evidence (Coelho et al., 2021; Fazli et al., 2021; Tran et al., 2020; Zhang et al., 2020), indicating that a range of health problems and complaints are associated with poor IEQ as depicted in Fig. 1 below. The problems range from temporary sensory irritation of the respiratory tract to diseases that can be life threatening.

1.1 Indoor Activity Time Trends

The occupancy of individuals in indoor environments should be investigated and associated with temporal variations of air contaminants to implement a comprehensive air quality management framework. Indoor occupants of different settings such as medical staff (medical doctors, nurses, receptionists) in hospitals, researchers in laboratories, and even children at schools tend to spend around 90% of their time indoors (Coelho et al., 2021; Dudzinska & Staszowska, 2021; Fazli et al., 2021; Ruiz-Jimenez et al., 2022). The characteristic of the temporal variation and occupancy of each sensitive indoor setting is the shift-based routines and the specific operational schedules associated with each setting. Typical patterns are not merely representative of the level of human presence or occupancy but can be used as initial point of investigation for IAQ strategies (Wi et al., 2021). Recognizing when and where occupants are most susceptible to prolonged exposure, this study aims to bridge the gap between temporal dynamics and effective IAQ interventions, ensuring that solutions align seamlessly with the distinct patterns of time spent indoors in critical environments (Tran et al., 2020; Weon et al., 2019). Figure 2 below illustrates the impact of poor air quality on health. It is observed that the most common problems were eye irritation, fatigue, coughing and irritation of respiratory airways and have been associated with bad air

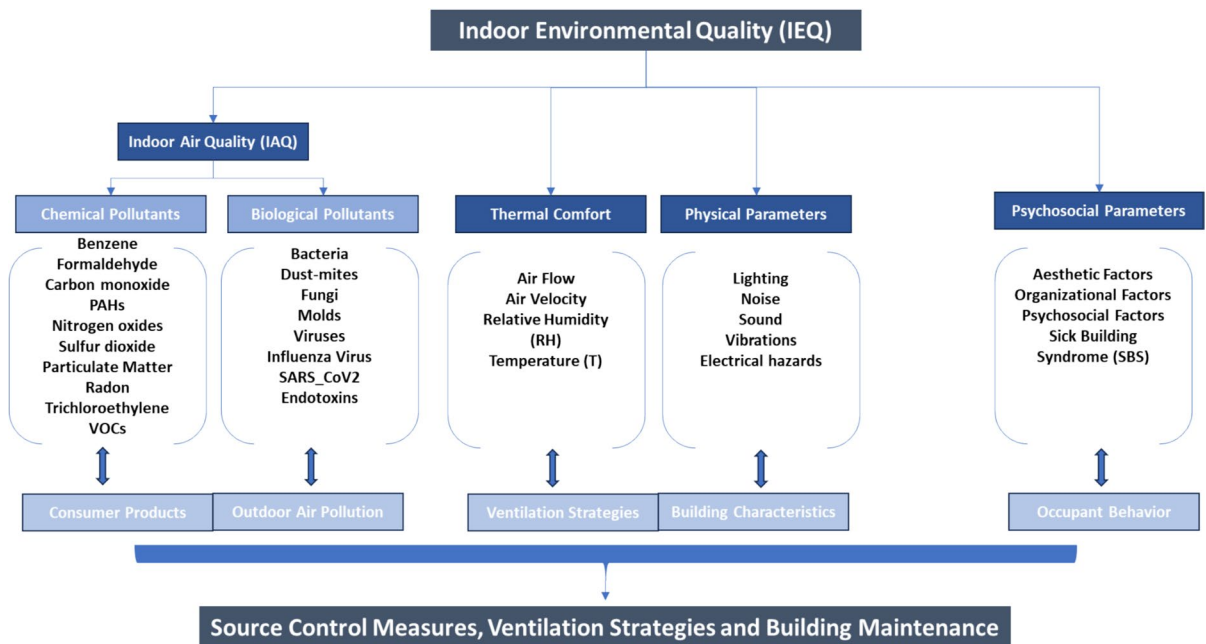


Fig. 1 Relation between Indoor Environmental Quality (IEQ) and Indoor Air Quality (IAQ) and its influencers' multifactor parameters

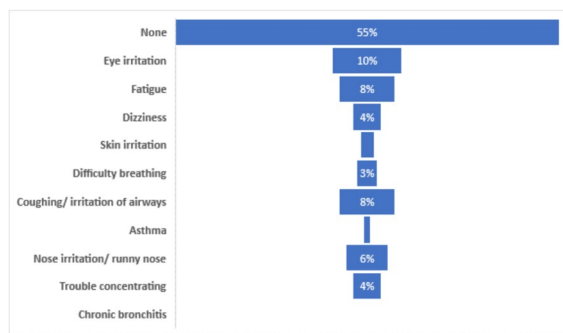


Fig. 2 Health symptoms reported by indoor occupants linked to IAQ. (Mendes et al., 2024)

quality (Chuanqi & Qianqian, 2020; Mendes et al., 2024).

1.2 Air Pollutants and Health Effects

Although the same air might be exchanged between indoor and outdoor settings, yet indoor air is different from ambient air in terms of contaminants' sources, types, levels, dispersion mechanisms, proximity of sources, emission rates, construction design and age,

and the ventilation systems in the building. Several air pollutants have been monitored and studied in peer-reviewed papers for their adverse effects on human health (Chen et al., 2022; Chuanqi & Qianqian, 2020; Cunha-Lopes et al., 2019; Fang et al., 2019; Faria et al., 2020; Gaio et al., 2019; World Health Organization (WHO), 2024; Liu et al., 2020; Martins et al., 2020; Maung et al., 2022; McGuinn et al., 2019; Mohd Isa et al., 2020; Othman et al., 2019; Pathak et al., 2020; Shetty et al., 2021; Xia et al., 2020). Common indoor air pollutants include particulate matter (TSP, PM₁₀ and PM_{2.5}), ozone (O₃), Radon, poly aromatic hydrocarbons (PAHs), NO and NO₂, SO₂, CO as well as a wide range of volatile organic compounds (VOCs) and bioaerosols (Chen et al., 2022; Chuanqi & Qianqian, 2020; Cunha-Lopes et al., 2019; Dudzinska & Staszowska, 2021; Fang et al., 2019; Faria et al., 2020; Gaio et al., 2019; World Health Organization, (WHO) 2024; Kelly & Fussell, 2019; Liu et al., 2020; Martins et al., 2020; Maung et al., 2022; McGuinn et al., 2019; Mohd Isa et al., 2020; Othman et al., 2019; Pathak et al., 2020; Shetty et al., 2021; Xia et al., 2020). In the aggregate of health effects, particulate matter (PM) can cause cardiovascular and respiratory diseases. Coarse particles (PM_{10-2.5}) when

inhaled can reach the upper parts of the lungs, pharynx and trachea (World Health Organization, (WHO) 2024; Maung et al., 2022) while fine particulate matter (PM_{2.5}) with less than 2.5 µm in diameter can penetrate deeply into the bronchi and alveoli regions of the lungs provoking lung cancer, inflammatory reactions (IR), oxidative stress (OS) and mutagenicity, and respiratory and cardiopulmonary diseases, increasing their harmful potential (World Health Organization, (WHO) 2024; Maung et al., 2022). Carbon monoxide (CO), an odourless and colourless gas generated by incomplete combustion, reduces the blood's ability to carry oxygen (Cunha-Lopes et al., 2019; Xia et al., 2020). It has an affinity for the oxygen carrying sites on the hemoglobin in the blood 200 times greater than oxygen, hence the higher the CO levels, the greater the displacement of oxygen occurs in tissues, and the more oxygen deficient the individual becomes. Major symptoms associated with CO exposure include shortness of breath, headaches, and nausea (Cunha-Lopes et al., 2019; Pathak et al., 2020; Xia et al., 2020).

Carbon dioxide (CO₂), on the other hand, is also an odourless and colourless product of carbon combustion. CO₂ levels are usually greater indoors than outdoor where the primary CO₂ source is human metabolism and breathing, gas-cooking appliances, space heaters, wood-burning appliances, and tobacco smoke as well as combustion by-products (automotive traffic), compressed CO₂ (fire extinguishers), dry ice, and aerosol propellants (Cunha-Lopes et al., 2019; Tran et al., 2020; Xia et al., 2020). Exposure to high CO₂ levels (> 1000 ppm) is an important risk factor of sick building syndrome (SBS) (Arikan et al., 2018; Pathak et al., 2020). Sulfur dioxide (SO₂) is generated in the atmosphere from coal-fired power plants, anthropogenic, mineral and volcanic sources (Chen et al., 2022; Shetty et al., 2021). It is corrosive, toxic and at high concentrations can cause life threatening pulmonary edema where coughing, shortness of breath, difficulty in breathing and tightness in the chest can be experienced (Chen et al., 2022; Pathak et al., 2020; Shetty et al., 2021). On the other hand, nitrogen oxides (NO_x) originate from vehicular, industrial and combustion emissions (Chen et al., 2022; Shetty et al., 2021), and may cause nausea, irritation in the eyes and nose, fluid forming in lungs and shortness of breath. At high levels, nitrogen oxides

may lead also to swelling of the throat, long-term asthma, and respiratory diseases (Chen et al., 2022; Shetty et al., 2021). VOCs are mainly released from housekeeping and maintenance products, building materials, industrial emissions, furnishings, equipment, pesticides, and insecticides [4, 20, 22]. Food manufacturing and industrial operations are equally known to generate organic chemicals that include VOCs, some of which are toxic, mutagenic and carcinogenic and can cause adverse health effects (Chuanqi & Qianqian, 2020; Fang et al., 2019; Liu et al., 2020). Typical symptoms of VOC exposure include headache, nausea, and irritation of the eyes, nose, and throat.

Contamination by indoor microorganisms and bio-aerosols is still a major health concern, particularly in developing countries (Liu et al., 2020). Health risks caused by contagious microorganisms and bio-aerosols including bacteria and viruses may include allergies, cardiovascular and respiratory infections (Martins et al., 2020). Fungi are widely distributed and pose a serious threat to household occupants due to ascomycota (a bag-like structure), and basidiomycota (club-shaped structure) (Liu et al., 2020; Martins et al., 2020; Xia et al., 2020). It is noteworthy to mention that biological contaminated agents are produced by animals, house dust, plants, and insects (Othman et al., 2019). The origin and health impacts of air contaminants and pollutants are summarized in Table 1 below.

Research shows that individuals exposed to indoor air pollutants have lower academic performance, less cognitive ability and productivity, lower problem-solving skills, absent-mindedness, and lower socio-economic behaviours (Mohd Isa et al., 2020; Pathak et al., 2020; Tran et al., 2020; Wi et al., 2021). In addition, trauma, depression, and anxiety have been shown to develop more frequently in individuals continuously exposed to poor IAQ compared to those living in environments with a better air quality (Shetty et al., 2021). Moreover, indoor air pollution affects the cardiovascular system and blood circulation as it increases the risk of heart diseases including heart failure (HF), hypertensive heart disease (HHD), pericardial disease (PD), rheumatic heart disease (RHD) (McGuinn et al., 2019), congenital heart disease (CHD), cardiomyopathy (Gaio et al., 2019), coronary artery disease (CAD), arrhythmia, congenital heart disease (CHD), and myocardial infarctions (MI)

Table 1 Origin and health impacts of air pollutants

Pollutant	Origin	Impact	Ref
PM ₁₀	Coarse particles are produced by mechanical processes such as erosion, grinding, coagulation or condensing. Coarse particles are rich in crustal matter that result from re-suspension of soils and surface dusts. They are also rich in sea salt (NaCl) and other naturally occurring earth constituents	PM ₁₀ can cause cardiovascular diseases, and has severe impacts on human respiratory system. They reach the upper parts of the lungs, pharynx and trachea. PM ₁₀ can also cause severe ecological effects and can lead to serious degradation of terrestrial and aquatic ecosystems through the process of scavenging and acid rain	World Health Organization (WHO), 2024; Maung et al., 2022; Martins et al., 2020; Xia et al., 2020; Othman et al., 2019)
PM _{2.5}	Fine particles which are less than 2.5 µm in diameter can stay in the atmosphere for days or weeks. They are mainly formed from chemical reactions, such as condensation of low volatile organics. They also contain more soluble inorganic components such as sulfate (SO ₄ ²⁻) and ammonium (NH ₄ ⁺)	PM _{2.5} is considered more dangerous and penetrates deeply into the bronchi and alveoli regions of the lungs provoking lung cancer and respiratory diseases	World Health Organization (WHO), 2024; Maung et al., 2022; Martins et al., 2020; Xia et al., 2020; Othman et al., 2019)
CO	CO is an odorless and colorless gas that is generated by incomplete burning of carbon-based fuels and from tobacco smoke	CO is dangerous to human health as it reduces the blood's ability to carry oxygen. It has an affinity for the oxygen carrying sites on the hemoglobin in the blood of 200 times greater than oxygen. As it displaces the oxygen, carbon monoxide prevents the distribution of the needed oxygen where the tissues become oxygen-deprived. As exposures increase, the individual may experience severe headaches, mental confusion, dizziness, nausea, rapid breathing, and fainting on mild exertion	Cunha-Lopes et al., 2019; Tran et al., 2020; Xia et al., 2020; Pathak et al., 2020; Arikkan et al., 2018)
CO ₂	CO ₂ is also an odorless and colorless product of carbon combustion. Common indoor sources may also be human metabolism and breathing, gas cooking appliances, space heaters, wood-burning appliances, automotive traffic and tobacco smoke	Exposure to high CO ₂ levels greater than 1000 ppm is an important risk factor of sick building syndrome (SBS)	Cunha-Lopes et al., 2019; Tran et al., 2020; Xia et al., 2020; Pathak et al., 2020; Arikkan et al., 2018)
SO ₂	SO ₂ is generated from coal-fired power plants, anthropogenic, mineral and volcanic sources	SO ₂ contributes to acid rain and aerosol formation as it acts as a precursor for sulfuric acid. It is also very corrosive and toxic and at high concentrations can cause life threatening pulmonary edema where coughing, shortness of breath, difficulty in breathing and tightness in the chest are mainly experienced	Pathak et al., 2020; Shetty et al., 2021; Chen et al., 2022)
NO/NO ₂	Vehicular, industrial and combustion emissions are considered the major sources of nitrogen oxides and particularly NO ₂	NO ₂ is a dangerous pollutant and a component of acid rain. This gas has many undesirable health effects on humans, including asthma and other cardiovascular and respiratory diseases resulting in swelling of oxygen pathways in the human body	Pathak et al., 2020; Shetty et al., 2021; Chen et al., 2022)

Table 1 (continued)

Pollutant	Origin	Impact	Ref
VOC	Volatile organic compounds (VOCs) are released from many housekeeping and maintenance products, building materials, industrial emissions, furnishings, equipment, pesticides and insecticides. Food manufacturing and industrial operations have been known to generate organic chemicals that include VOCs	VOCs cause adverse health effects and some of which are toxic, mutagenic and carcinogenic. The symptoms of VOCs may cause slight irritation including headache, nausea, and irritation of the eyes, nose, and throat. High toxicity levels eventually may lead to death	Chuanqi & Qianqian, 2020; Fang et al., 2019; Liu et al., 2020
Bioaerosols	Bioaerosols can be fungi, bacteria, pathogens and viruses. Bacteria include Archaea, mycobacteria, Gram positive and negative bacteria. Archaea are methanogens microorganisms that have unique membrane lipids, a cell wall devoid of peptidoglycan, intrinsic capacity to resist antibiotics. Non-tuberculous mycobacteria (NTM) are mainly found in water related sources, soil and metal-working fluids. Naturally occurring Mycobacterium tuberculosis in aerosols can be viable but may not be culturable. Molecular methods have revealed that fecal anaerobes are major constituents of bioaerosols in various environments such as agricultural settings. Given their complex structures, viruses act as bioaerosol agents that are difficult to culture. Airborne viruses are markedly absent in culture media where there is no universal assay for viruses	Bacteria such as Archaea can induce chronic inflammation in the lungs and lead to sensitization diseases. NTM are microorganisms that can cause lung diseases whether or not they are infectious. Whereas Gram positive and negative bacteria have potential impact on respiratory tract. Viruses on the other hand include transmitted viral diseases such as influenza A and B (flu), coronavirus (COVID-19, common cold and severe acute respiratory syndrome—SARS), adenovirus (lung infections and common cold), norovirus (gastrointestinal illnesses), and morbillivirus (measles and mumps). All viruses can induce inflammatory response	Liu et al., 2020; Martins et al., 2020; Xia et al., 2020; Othman et al., 2019)

(Zhang et al., 2021). A summary of the latest research studies about the health effects of indoor air pollutants is described in Table 2.

An analysis of the reviewed studies in Table 2 reveals a consistent association between poor indoor air quality and a range of adverse health outcomes across diverse indoor environments and population groups. Commonly measured pollutants include PM_{2.5}, VOCs, total VOCs TVOCs, NO₂, O₃, CO₂, and aldehydes such as formaldehyde. Other hazardous substances such as radon, synthetic phenolic antioxidants, and flame retardants were also assessed, alongside environmental parameters including temperature and relative humidity. The main health effects reported in these studies range from respiratory symptoms and cardiovascular changes to more systemic issues like fatigue, sleep disturbances, allergies, and sick building syndrome (SBS). Vulnerable populations, such as children, pregnant women, and elderly were the targeted occupants and the main focus of these investigations, underscoring the heightened sensitivity of certain demographic groups to indoor air pollution. This review highlights the importance of comprehensive, population specific IAQ assessments to inform effective control strategies and protect occupant health.

Table 3 provides an overview of indoor air contaminants commonly encountered in residential and commercial settings, categorized into microbial (Chegini et al., 2020; Chmiel et al., 2019; Marcu et al., 2021; Stanaszek-Tomal, 2020), chemical (Domínguez-Amarillo et al., 2020; Majd et al., 2019; Wan Mahiyuddin et al., 2023), particulate (Bennett et al., 2019), and gaseous contaminants (Zhang & Srinivasan, 2020). Microbial contaminants, including bacteria, viruses, and fungi/Molds, originate from various sources such as human occupants' metabolism, ventilation systems, and damp conditions of high relative humidity (RH). These contaminants can lead to respiratory infections, exacerbation of asthma and respiratory diseases along with allergic reactions. Chemical contaminants like CH₂O, C₆H₆, NO₂, and O₃ stem from building materials, furniture, tobacco smoke, and combustion processes (Bennett et al., 2019; Chegini et al., 2020; Chmiel et al., 2019; Domínguez-Amarillo et al., 2020; Kelly & Fussell, 2019; Majd et al., 2019; Marcu et al., 2021; Stanaszek-Tomal, 2020; Zhang & Srinivasan, 2020). PM₁₀ and PM_{2.5} is released through activities like cooking, smoking, and

construction, while gaseous pollutants such as carbon dioxide and radon stem from human respiration, poor ventilation, soil, and certain building materials.

Table 4 provides a comprehensive breakdown of the standard limits of air quality indicators crucial for assessing IAQ within residential and commercial buildings. Each parameter plays an important and essential role in determining the overall health, comfort, and well-being of occupants within these confined spaces. The standards and acceptable levels of IAQ indicators set by international organizations like WHO, Environmental Protection Agency (EPA), National Ambient Air Quality Standards (NAAQS) and The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) is depicted in Table 5 below (Awad & Jung, 2021; Jung & Awad, 2021; Mannan & Al-Ghamdi, 2021; Sabbagh & Alama, 2021). Therefore, it is important to compare any levels of contaminants that are measured with their standard limits to assess the quality of indoor air in any confined space.

1.3 Purpose of Research

Historically, public concern with IAQ has evolved as pollutants from various emission sources were identified in indoor environments at levels associated with adverse health effects (Awad & Jung, 2021; Baudet et al., 2021; Brambilla & Capolongo, 2019; Fazli et al., 2021; Hassan & Zeeshan, 2022; Hiwar, et al., 2021; Kenarkoohi, et al., 2020; Odone et al., 2019; Pantelic et al., 2023; Ruiz-Jimenez et al., 2022; Taushiba et al., 2023; Vonci et al., 2019). This is particularly true in sensitive environments such as schools and hospitals, where poor air quality could negate the purpose of the stay leading to exacerbate health risks (Awad & Jung, 2021; Baudet et al., 2021; Brambilla & Capolongo, 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Kenarkoohi et al., 2020; Odone et al., 2019; Riveron et al., 2023; Taushiba et al., 2023; Vonci et al., 2019). While many recent studies (Ali et al., 2021; Amoatey et al., 2020; Cheek et al., 2021; Chen et al., 2020; Facciola et al., 2019; Gola et al., 2019; Liu et al., 2020; Mentese et al., 2020; Reis et al., 2021; Sakellaris et al., 2020; Szabados et al., 2021; Uscinowicz et al., 2023; Zeng et al., 2023; Zhang et al., 2023a) have focused on air quality in various indoor environments, limited work examined IAQ in certain sensitive environments.

Table 2 A summary of the latest research studies about the health effects of indoor air pollutants

Ref	Location	Number of samples	Year	Study objectives	Health diseases and symptoms Monitored	Pollutants measured
Chen et al. (2020)	Taipei, Taiwan	100 elderly people over 65 years of age	2020	The study aimed to explore indoor air pollution, blood pressure, and cardiovascular effects among elderly people	Heart rate (HR) and blood pressure (BP)	PM _{2.5} , TVOCs, temperature, and relative humidity
Sakellaris et al. (2020)	Finland, France, Greece, Hungary, Italy, the Netherlands, Portugal, Spain	1299 participants from 37 office buildings	2020	This study investigated the relationship between building occupants' health symptoms and indoor pollutant concentrations with 1299 participants supported by the European OFFICAIR research	Headache, fatigue, eye irritation, respiratory, cardiac, skin, and general symptoms	VOCs, aldehydes, ozone, and NO ₂
Cheek et al. (2021)	China, USA, Canada, Denmark and Mongolia	NA	2021	A systematic literature review examined the effects of portable air purifiers (PAPs) on indoor air quality (PM _{2.5}) and health implications	BP, respiratory parameters, and pregnancy	PM _{2.5}
Mentes et al. (2020)	Çanakkal, Turkey	121 homes	2020	This study investigated IAQ, pollution sources, the frequency of SBS, and respiratory health symptoms through a long-term monitoring program	sick building syndrome (SBS), and respiratory health indicators	PM, VOCs, bioaerosols, CO ₂ , benzene, Cladosporium sphaerospermum, Penicillium spp., air temperature, and humidity
dos Reis et al. (2021)	Portugal	NA	2021	The study purposed to fill the health gap and reduce pulmonary diseases due to poor IAQ	pulmonary diseases and allergies	VOCs, PM _{2.5}
Szabados et al. (2021)	Italy, Poland, Slovenia, Czech Republic, and Hungary,	64 primary school buildings	2021	This study explored the relationship between IAQ and associated health risks in primary school buildings	Carcinogenic and Non-carcinogenic risks	VOCs, aldehydes, PM _{2.5} , CO ₂ , radon (Rn)

Table 2 (continued)

Ref	Location	Number of samples	Year	Study objectives	Health diseases and symptoms Monitored	Pollutants measured
Amoatey et al. (2020)	The Gulf Cooperation Council (GCC) countries except Kuwait, the United Arab Emirates (UAE), and Kingdom of Saudi Arabia	NA	2020	The study highlighted the relationship between human health and IAQ in the Gulf Cooperation Council (GCC) countries	respiratory symptoms and SBS	CO ₂ , VOCs, siloxanes, flame retardants, synthetic phenolic antioxidants
Zeng et al. (2023)	Hong Kong	9 typical household	2023	This study evaluated the role of indoor gas pollution and associated health risks	chronic non-carcinogenic and carcinogenic risks, and respiratory symptoms,	Acrolein, nitrogen dioxide (NO ₂), HCHO, benzene, acetaldehyde, ethylbenzene and dichlorobenzene
Pantelic et al. (2023)	USA	14 participants	2023	The study evaluated the effects of IAQ on cardiopulmonary health, environmental comfort, and sleep quality	Cardiopulmonary disease	PM _{2.5} , CO ₂ , TVOCs, Relative Humidity, and temperature

Therefore, the objective of this review paper is to assess and discuss the impact of harmful pollutants on IAQ in vulnerable indoor environments, including healthcare facilities, residential homes, and educational institutions. It seeks to evaluate key factors influencing IAQ parameters. By synthesizing findings from existing literature and incorporating empirical data from various building types, the study enhances the understanding of IAQ challenges and provide practical insights on maintaining optimal IAQ standards. The goal is to support the promotion of occupational health and safety, particularly among the most vulnerable populations. In one aspect, analysing IAQ in a variety of old and historic buildings is of utmost importance to determine all possible internal pollution factors that are responsible for the negative health effects in certain types of buildings, particularly residential and commercial ones. On the other hand, this review provides a framework for the design and evaluation of introducing various new components into the building's interior environment for a healthier quality of life and better indoor environmental quality (IEQ). This study will guide policymakers, healthcare professionals, construction engineers, and facility managers in developing effective strategies to mitigate indoor air pollution, enhance indoor air quality (IAQ) and improve the overall indoor environmental quality (IEQ).

2 Methodology

This review study provides a comprehensive analysis in a qualitative approach through findings obtained from a wide range of peer-reviewed articles from leading and top journals and databases most of which are in first (Q₁) and second quartile (Q₂) in public health, environmental health, air quality, and building sciences research domains. Such study ensures a comprehensive and up-to-date analysis of indoor air pollutants and their health effects. In addition, this study draws upon a diverse body of literature sourced from global studies on residential settings, covering a broad spectrum of indoor air pollutants and their control/mitigation programs.

The use of keywords and logical operators were an essential and crucial part of our research strategy. The search strategy involved a combination of free-text keywords and Boolean operators (AND,

Table 3 Sources of indoor air pollutants (IAP)

Pollutants	Sources	Ref
Microbial pollutants		
- Bacteria	Human occupants, ventilation systems, pet waste, septic systems, agriculture, wildlife, and sanitary sewage overflows	Chegini et al., 2020; Marcu et al., 2021)
- Viruses	Infected individuals, poor ventilation, and tiny airborne particles	Chmiel et al., 2019)
- Fungi/Mold	Damp conditions, water leaks, poor ventilation	Stanaszek-Tomal, 2020)
Chemical pollutants		
- Formaldehyde (CH ₂ O)	Building materials, furniture, tobacco smoke	Domínguez-Amarillo et al., 2020)
- Benzene (C ₆ H ₆)	Tobacco smoke, paints, synthetic materials	Wan Mahiyuddin et al., 2023)
- Nitrogen Dioxide (NO ₂)	Combustion processes, gas appliances	Majd et al., 2019)
- Ozone (O ₃)	Electronic devices, printers, photocopiers, and air cleaners that emit ozone as a by-product	Majd et al., 2019)
Particulate Matter		
- PM ₁₀ & PM _{2.5}	Cooking, candles, tobacco smoke, motor combustion Construction activities, dust	Bennett et al., 2019)
Gaseous pollutants		
- Carbon Dioxide (CO ₂)	Human respiration, inadequate ventilation	Zhang & Srinivasan, 2020)
- Radon	Soil, building materials	Zhang & Srinivasan, 2020)

Table 4 Standard limits of different air quality indicators

Parameter	Description	Measurement	Acceptable Levels	Ref
Particulate Matter (PM)	Coarse and fine particles suspended in the air	Micrograms per cubic meter (µg/m ³)	PM _{2.5} : < 10 µg/m ³ PM ₁₀ : < 20 µg/m ³	Parmaksız et al., 2021; Cocârță, et al., 2021)
Volatile Organic Compounds (VOCs)	Organic chemicals emitted as gases from certain solids or liquids	Parts per billion (ppb)	< 500 ppb	Kraus & Šenitková, 2020)
Carbon Dioxide (CO ₂)	Colourless, odourless gas produced by human respiration and combustion	Parts per million (ppm)	< 1000 ppm	Licina & Langer, 2021)
Formaldehyde (HCHO)	Colourless, strong-smelling chemical used in building materials and household products	Parts per billion (ppb)	< 50 ppb	Dehghani, et al., 2020)
Radon (Rn)	Naturally occurring radioactive gas that can seep into buildings from the ground	Picocuries per litre (pCi/L)	< 4 pCi/L	Rojas, et al., 2024)
Humidity	Amount of water vapour in the air	Percentage (%)	30%—50% (for comfort)	Demaneaga, et al., 2021)
Temperature	Degree of hotness or coldness in the air	Celsius (°C)	20 °C—25 °C (for comfort)	Schütze & Sauerwald, 2020; Salthammer & Morrison, 2022; Mkoombindo, et al., 2022)
Air Exchange Rate (ACH)	Number of times the entire volume of air in a space is replaced per hour	Air changes per hour (ACH)	0.5—1.5 ACH	Tran et al., 2020)

Table 5 The standard and acceptable levels of IAQ indicators set by international organizations

Pollutants	Saudia Arabia (2021)	United Arab Emirates (UAE) (2021)	WHO (2021)	United States (2021)
HCHO	N/A	0.08 ppm (8 h average)	0.2 mg/m ³ (long term)	0.1 ppm (ASHRAE)
TSP (Total suspended particles)	340 µg/m ³ (Saudi Standard)	150 µg/m ³ (8 h average)	N/A	25 µg/m ³ (24 h)
NO ₂	N/A	N/A	200 µg/m ³ (1 h)	0.053 ppm (NAAQS)
CO	35 ppm (Saudi Standard)	N/A	51 ppm (30 min average)	25 ppm (EPA) (8 h)
VOCs	N/A	300 µg/m ³ (8 h average)	N/A	N/A

OR and NOT) to identify relevant literature and restrict the search down to those keywords that had no overlap between the below stated terms. Keywords and key search terms included “Indoor Air Quality,” “Critical Indoor Environments,” “Health Impacts,” “Indoor Air Contaminants,” “Indoor Climate and Settings,” “IAQ in Hospitals,” “IAQ in Schools,” “IAQ in Residential Buildings,” “IAQ in Office Spaces,” “Ventilation system,” “Occupational Health & Safety” and “Occupant Well-being”. To ensure this review reflects recent findings and remains up-to-date, the focus was placed on literature published between 2019 and 2024. Peer-reviewed research studies and literature reviews were preferred for consideration, whereas conference abstracts and non-scholarly sources were not included to preserve the scientific comprehensive approach of our analysis. Data extracted from eligible studies included information on pollutant types, sources, exposure levels, study design, targeted populations, health outcomes, and any proposed mitigation or pollution control strategies. Studies conducted solely on animals or laboratory models without direct relevance to human indoor environments were excluded. No restrictions were applied regarding geographical location. Full-text articles were then assessed against inclusion criteria, which required that studies focus on IAQ or indoor air pollutants, evaluate associated health effects/risks and mitigation strategies, and provide data relevant to critical building environments. The extracted data were then synthesized thematically to identify key findings, emerging trends, and research gaps in the literature. By employing a thoughtful combination of keywords, Boolean operators, and adherence to database-specific controlled vocabularies, this review study aimed to cast a wide pool of

contaminants that are considered harmful to human health, thereby enriching the scope and depth of our comprehensive review.

3 Results and Discussion

3.1 Indoor Air Quality in Hospitals

This section describes the latest research on chemical and microbiological contamination of healthcare facilities in various peer-reviewed journals over the past five years particularly between 2019 and 2024. Indoor air quality is highly affected by ambient emission sources particularly in congested urban areas invariably associated with vehicle-induced emissions as well as construction and industrial emissions. IAQ in certain sensitive environments such as hospitals is a critical factor for its occupants that could negate the purpose of the visit if IAQ deteriorates. Although many research articles have studied IAQ in schools, residential, and office buildings, the field of healthcare facilities has been overlooked (Ali et al., 2021; Baudet et al., 2021; Brambilla & Capolongo, 2019; Facciola et al., 2019; Gola et al., 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Kenarkoohi et al., 2020; Liu et al., 2020; Odone et al., 2019; Riveron et al., 2023; Taushiba et al., 2023; Uscinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a). Although healthcare facilities, treating the most vulnerable populations, should have optimal IAQ, however studies show that, those, including hospitals, clinics and outpatient centers, primary health care centers, long-term care facilities, and others have of airborne pathogens, VOCs, and other indoor pollutants that can lead to adverse patient outcomes and indoor occupants’ health challenges (Baudet et al., 2021; Brambilla & Capolongo,

2019; Facciola et al., 2019; Gola et al., 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Kenarkoochi et al., 2020; Odone et al., 2019; Riveron et al., 2023; Taushiba et al., 2023; Uścinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a).

The impact of air quality on patient recovery rates and healthcare worker safety and productivity in healthcare facilities has been studied and Table 6 provides a summary of the most recent studies describing various pollutants measured in hospital indoor environments. The reviewed studies provide a multidimensional and comprehensive understanding of IAQ in diverse healthcare settings across Pakistan, France, India, the UK, China, and Lebanon. They collectively emphasize the complexity of indoor pollutants, their variability across contexts, and the associated health implications, particularly in vulnerable environments like hospitals. The results of studies of IAQ in hospitals are presented in this section however major gaps and limitations are also highlighted and depicted in Table 6. Three studies discussed Microbial pollutants, such as bacteria and fungi (Baudet et al., 2021; Hassan & Zeeshan, 2022; Taushiba et al., 2023). In Pakistan they found *Staphylococcus spp.* (53%) and *Aspergillus spp.* (67%) dominant in hospitals with varying disinfection protocols and ventilation systems (Hassan & Zeeshan, 2022), while in France, they observed moderate bacterial (14 CFU/m³) and fungal (7 CFU/m³) counts, with dominant species being *Staphylococcus*, *Micrococcus*, and *Penicillium* (Baudet et al., 2021). In India extremely high microbial loads, with bacterial concentrations reaching 1389.21 CFU/m³ and fungi at 786.34 CFU/m³, were reported indicating poor IAQ and possibly inadequate air control in those facilities (Taushiba et al., 2023). In terms of particulate matter, Taushiba et al. found alarming PM_{2.5} levels indoors (up to 149.41 µg/m³), suggesting substantial indoor air pollution likely attributed to infiltration from outdoor sources and internal activities (Taushiba et al., 2023), while Chamseddine et al. found that PM_{2.5} and PM₁₀ levels in hospitals exceeded air quality guidelines by 2–3.5 times during warm seasons, with strong indoor-outdoor correlations indicating that regional events like dust storms significantly influence IAQ, especially in poorly insulated and ventilated facilities (Chamseddine et al., 2019). In terms of Volatile Organic Compounds (Baudet et al., 2021; Riveron et al., 2023; Zhang et al., 2023a), VOCs were identified as major indoor pollutants in hospitals

across studies from France, the UK, and China. High concentrations of VOCs such as 2-propanol, acetone, formaldehyde, and benzene were detected, primarily linked to medical procedures and external emissions. Notably, elevated levels of formaldehyde and benzene were associated with an increased risk of recurrent asthma in children, underscoring the health risks posed by VOC exposure in healthcare settings. Only one study (Zhang et al., 2023a) explicitly explored health outcomes, showing a significant association between indoor air pollutants and childhood asthma recurrence. Passive smoking, formaldehyde, and benzene were identified as key risk factors, underlining the critical role of IAQ in respiratory health.

The reviewed studies reinforce the urgent need for regular and continuous monitoring of IAQ in hospital environments to ensure the safety and well-being of both patients and healthcare workers. Moreover, such studies highlight the importance of improving ventilation systems to reduce pollutant accumulation and promote healthier indoor air quality. Additionally, the findings suggest a shift toward the use of low-emission materials and procedures in healthcare settings to minimize the introduction of harmful substances. Lastly, there is a clear call for more comprehensive, human-centered research that directly links specific indoor pollutant exposures to health outcomes, enabling the development of evidence-based policies and interventions.

The standards and requirements for hospitals must accommodate for the comfort of a wide range of occupants with an equally wide range of vulnerabilities (Baudet et al., 2021; Brambilla & Capolongo, 2019; Chamseddine et al., 2019; Facciola et al., 2019; Gola et al., 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Kenarkoochi et al., 2020; Odone et al., 2019; Riveron et al., 2023; Taushiba et al., 2023; Uścinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a). Hospitals should be rated as high-performance buildings in terms of environmental sustainability and air quality to enhance and retain patients' healing process as well as working staff efficiency (Reis et al., 2021). Life threatening nosocomial infections could be acquired in hospitals as a result of poor IAQ, and patients are at risk of developing infections when hospitals' indoor environment is affected by various indoor and outdoor sources of air contaminants. In hospitals, the time spent indoors depends on the health condition of patients, allocated

Table 6 A summary of research studies on IAQ in hospitals

Ref	Location	Number of samples	Pollutants Measured	Results	Gaps and Limitations
Hasan and Zeeshan (2022)	Pakistan	16 sampling sites were selected from two public hospital buildings	The objective was to compare the concentration, composition, and prevalence of microbial organisms in hospital spaces with different VS, disinfection methods, and occupancy levels	Bacterial microbiota – <i>Bacillus</i> spp. (11%) – <i>Staphylococcus</i> spp. (53%) – <i>Micrococcus</i> spp. (30%) Fungal microbiota – <i>Penicillium</i> spp. (28%) – <i>Aspergillus</i> spp. (67%)	– Spatial variation of microbial contaminants was studied however, this study didn't explore seasonal variation of microbial pollutants – This study didn't explore chemical pollutants concentrations such as CO, NO_x, SO₂, PM₁₀, PM_{2.5} and VOCs and didn't explore any indoor-outdoor (IO) correlation of pollutants – This study didn't introduce any air quality management framework to mitigate the high concentrations of microbial pollutants
Baudet et al. (2021)	France	6 private healthcare facilities	This study determined chemicals and microorganisms in health care and care facilities	Chemical compounds (µg/m ³) – aldehydes (formaldehyde: 11.4 and acetaldehyde: 6.5) – terpenes (limonene: 4.3) – alcohols (ethanol: 378.9 and isopropanol: 23.6) – ketones (acetone: 18.8) Bacterial microbiota – <i>Staphylococcus</i>, <i>Micrococcus</i> and <i>Bacillus</i> genera = 14 cfu/m³ Fungal microbiota – <i>Cladosporium</i> and <i>Penicillium</i> = 7 cfu/m³ Particulate Matter (PM) – PM_{2.5} = 9.0 µg/m³	– Spatial variation of microbial and chemical contaminants was studied however, this study didn't explore seasonal variation of chemical and microbial pollutants – This study didn't monitor gaseous chemical pollutants such as CO, NO_x, SO₂, and VOCs and didn't explore any indoor-outdoor (IO) correlation of pollutants – This study didn't introduce any air quality management framework to mitigate the high concentrations of chemical and microbial pollutants

Table 6 (continued)

Ref	Location	Number of samples	Pollutants Measured	Results	Gaps and Limitations
Taushiba et al. (2023)	India	3 different hospitals in Lucknow, India	This study assessed and identified indoor and outdoor air quality concentrations of PM _{2.5} , sub-micron particles, heavy metals, and microbial pollutants in three different hospitals in India particularly in Lucknow city	Particulate matter (PM) - PM_{2.5} = The highest indoor and outdoor concentrations obtained are 149.41 µg/m³ and 227.45 µg/m³ in hospital A and hospital C - Bacterial concentration was highest in hospital B with 1389.21 CFU/m³ - Fungi concentration was highest in hospital C with 786.34 CFU/m³	- Spatial variation of microbial contaminants and PM concentrations was studied however, this study didn't explore seasonal variation of chemical and microbial pollutants - This study didn't monitor gaseous chemical pollutants such as CO, NO_x, SO₂, and VOCs and didn't explore any indoor-outdoor (IO) correlation of pollutants - This study didn't introduce any air quality management framework to mitigate the high concentrations of PM pollutants and bacterial and fungi contaminants
Riveron et al. (2023)	United Kingdom	Over 600 air samples were collected over 31 months during the years of 2017 to 2020 at two hospital sites in Leicester, UK	This study determined the health risks of VOCs in hospitals' indoor air	The most abundant VOCs were 2-propanol (696.6 µg/m³), ethyl chloride (436.5 µg/m³), acetone (83.9 µg/m³), and hexane (58.5 µg/m³). The study concluded that the VOCs with the highest concentrations were most likely released through healthcare activities or the entrance of vehicle emissions	- This study didn't monitor biological pollutants and didn't explore and measure dust particles PM₁₀ and PM_{2.5} and other gaseous chemical pollutants such as CO, NO_x, and SO₂ - This study didn't assess any indoor-outdoor (IO) correlation of pollutants - This study didn't introduce any air quality management framework to mitigate the high concentrations of chemical pollutants such as VOCs

Table 6 (continued)

Ref	Location	Number of samples	Pollutants Measured	Results	Gaps and Limitations
Zhang et al. (2023a)	China	Asthma case (201 children with recurrent asthma) and control cohorts (242 healthy subjects)	This study examined the indoor environment of recurrent childhood asthma in China	Results showed that recurrent asthma was associated with passive smoking [aOR2.115 (95%-CI 1.275–3.508)]. Increased formaldehyde and benzene concentrations were strongly evident between asthmatic versus control groups and were strongly associated with an increased risk of asthma recurrence	– This study didn't monitor biological pollutants and didn't explore and measure dust particles PM₁₀ and PM_{2.5} and other gaseous chemical pollutants such as CO, NO_x, and SO₂ in order to determine their effects on asthma – This study didn't assess any indoor- outdoor (IO) correlation of pollutants. Only indoor VOCs (formaldehyde and benzene concentrations were correlated with increased risk of asthma recurrence) – This study didn't introduce any air quality management framework to mitigate the high concentrations of chemical pollutants such as VOCs particularly formaldehyde and benzene

Table 6 (continued)

Ref	Location	Number of samples	Pollutants Measured	Results	Gaps and Limitations
Chamseddine et al. (2019)	Lebanon	Around 374 samples were collected from 12 working areas of three Hospitals in Beirut, Lebanon during warm and cold seasons	This study involved a seasonal exposure assessment in a hospital environment using several air quality indicators including carbon monoxide (CO), carbon dioxide (CO₂), particulate matter (PM₁₀ and PM_{2.5}), and total volatile organic compounds (TVOCs). The distribution of and variation in the indoor and outdoor pollutant concentrations in 12 working areas across three hospitals were examined, with an emphasis on capturing seasonal variations	Results indicated that: - While indoor and outdoor CO levels were below air quality standards/guidelines, measured PM_{2.5} and PM₁₀ concentrations at several locations exceeded the standards 2 to 3.5 folds - Higher indoor PM levels during the warm season were recorded, particularly during regional desert storm events - The ingress of particles from the outdoor to indoor environment was evident with high correlations between indoor and outdoor PM_{2.5} (r between 0.83 and 0.92) and PM₁₀ (r between 0.74 and 0.86) levels, particularly during the warm season - Indoor to outdoor (I/O) ratios of PM_{2.5} and PM₁₀ were mostly < 1. In contrast, indoor levels of CO₂ and TVOCs exceeded outdoor concentrations during both the warm and cold seasons with I/O ratios > 1 across all sampling locations	Although this study has explored spatial and seasonal variation of chemical pollutants however it study didn't monitor biological pollutants and didn't measure microbial contaminants including bacteria viruses and fungi

time for visitors, and the working shifts of the staff rendering hospital environments special with a high occupancy ratio continuously. Therefore, maintaining high standards that reflect the requirements and expectations of indoor occupants is a continuous concern for healthcare providers at hospitals. The reviewed studies reinforce the urgent need for regular monitoring of IAQ in healthcare environments to ensure the safety and well-being of both patients and healthcare workers. They highlight the importance of improving ventilation systems to reduce pollutant accumulation and promote healthier indoor conditions. Additionally, there should be a shift toward the use of low-emission materials and procedures to minimize the introduction of harmful substances. Lastly, there is a clear call for more comprehensive, human-centered research that directly links specific indoor pollutant exposures to health outcomes, enabling the development of evidence-based policies and interventions.

3.2 Indoor Air Quality in Residential Buildings

Air pollution in urban areas is associated with a range of health effects, both respiratory and cardiovascular (Amoatey et al., 2020; Chamseddine et al., 2019; Cheek et al., 2021; Chegini et al., 2020; Chen et al., 2020; Kausar et al., 2023; Mentese et al., 2020; Pantelic et al., 2023; Reis et al., 2021; Sakellaris et al., 2020; Szabados et al., 2021; Zeng et al., 2023), along with increased cancer risk (Liu et al., 2020; Martins et al., 2020; Mohd Isa et al., 2020; Othman et al., 2019; Pathak et al., 2020; Xia et al., 2020). The global burden of disease classifies air pollution as the fourth leading cause of mortality globally; this is mainly attributed to the joint effects of household and ambient air pollution (Ali et al., 2021; Chamseddine et al., 2019; Liu et al., 2020; Martins et al., 2020; Mohd Isa et al., 2020; Othman et al., 2019; Pathak et al., 2020; Xia et al., 2020). Exposure may occur from multiple indoor and outdoor sources and is a characteristic of both human activities and the natural environment (Liu et al., 2020; Martins et al., 2020; Mohd Isa et al., 2020; Othman et al., 2019; Pathak et al., 2020; Xia et al., 2020). Worldwide, concerns regarding poor IAQ have continuously increased as pollutants from various sources are identified indoors at concentrations exceeding health thresholds. IAQ in residential buildings and houses is often deteriorated

by chemical substances emitted from heating, building materials, household products. The studies presented in Table 7 reflect IAQ assessments across diverse residential environments in the UAE, China, Saudi Arabia, and Pakistan, involving different ventilation systems, seasons, and pollutant types. Muhammad, et al. (Ali et al., 2021) discussed the use of solid fuel (SF), its harmful practices, and particularly the diseases caused by long-term inhalation of such fuel. Most of the pollutants produced with SF include PM, nitrous dioxide (NO₂), carbon monoxide (CO), and sulphur dioxide (SO₂) that may cause cardiovascular and pulmonary diseases including lung cancer (Ali et al., 2021; Liu et al., 2020). The indoor air pollution from household SF can affect vulnerable occupants such as elderly people, children, and women, resulting in symptoms such as shortness of breath, fatigue, drowsiness, and weight loss (Ali et al., 2021; Liu et al., 2020). Moreover, studies indicated that these indoor pollutants could cause and stimulate many toxic effects in the human body, including oxidative stress (OS), gene activation, and DNA methylation (Ali et al., 2021), which can lead to serious health problem if left untreated. In 2020, another study by Liu et al. (Liu et al., 2020) assessed IAQ and indoor temperature during the summer season in a kitchen in China. The comprehensive analysis of this study demonstrated that the amount of TVOCs and PM_{2.5} that were emitted in the kitchen exceeded the standard levels of Chinese guidelines (Liu et al., 2020). The study proposed installing a new ventilation system in the kitchens to enhance IAQ and thermal comfort within the closed environments (Ali et al., 2021; Liu et al., 2020). Kausar et al. (Kausar et al., 2023) targeted several indoor air pollutants that are mostly generated by household energy sources such as wood, coal, and tobacco. It was concluded that a significant relationship was found between IAQ and household energy sources (Kausar et al., 2023). This study revealed that indoor smoking causes high levels of PM_{2.5} ~ 1800 µg/m³, resulting in chronic obstructive pulmonary disease (COPD), asthma and lung cancer, mostly affecting vulnerable individual such as young children and women (Kausar et al., 2023; Wang et al., 2021). The study also stated that poor IAQ is linked to reduced cognitive performance, negatively affecting learning, creativity, and productivity. Another important cause for poor IAQ is indoor smoking is one of the main causes of human death worldwide, and its

Table 7 A summary of recent research studies on IAQ in residential buildings

Ref	Research location	Quantity of sample	Ventilation system	Season	Pollutants Measured
Awad and Jung (2021)	The Greens in Dubai, United Arab Emirates	20 residential houses	Naturally ventilated	Winter	- Temperature & Relative Humidity (RH) - CO - CO ₂ - CH ₂ O - VOCs - Total Suspended Particles (TSP) TVOC, CO, and O ₃ , CO ₂ , PM _{2.5} , HCHO
Sui et al. (2021)	Xi'an, China	1 residential building	- Natural infiltration - Window-opening ventilation - Two mechanical ventilation modes with low and high ventilation rates	Winter	
Deng et al. (2021)	Kunming, China	30 dwellings	Naturally ventilated	Spring, Summer, Autumn, and Winter	Temperature, relative humidity, PM _{2.5} , CO ₂ , benzene, toluene, HCHO, and xylenes-VOCs
Liu et al. (2020)	Changsha, China	16 cooks participated in this study	NA	Summer	CO, CO ₂ , TVOCs, and PM _{2.5} , and temperature
Alazazmeh and Mohammad Asif (2021)	Dammam, Saudia Arabia	No. of Floor = 2 Number of Occupants = 250	- Conventional packaged air conditioning unit (PACU) with constant air volume (CAV)	All seasons	Temperature, relative humidity, illumination, noise control, air movement, odor, and pollutants such as total suspended particulate (TSP), PM ₁₀ , SO ₂ , NO ₂ , CO, and VOCs
Kausar et al. (2023)	Pakistan	NA	NA	All seasons	PM _{2.5} , CO ₂ , CO, NO ₂ , polycyclic aromatic hydrocarbons (PAHs), HCHO, and airborne particles

harmful effects have been well described in various studies (Lim et al., 2021; Moreno-Rangel et al., 2021, 2023; Ni et al., 2020). Ni, Shi, and Qu et al. (Ni et al., 2020) investigated the relationship between indoor $PM_{2.5}$ and various lung diseases including chronic obstructive pulmonary disease (COPD), tuberculosis (TB), lung cancer, asthma, and interstitial lung disease (ILD). It is evident that both tobacco smoke and $PM_{2.5}$ exposures are considered risk factors for lung cancer and asthma (Lim et al., 2021; Ni et al., 2020).

Another research has been conducted to investigate IAQ and haze/dust storm events in a residential apartment in China (Sui et al., 2021), and concluded that compared to MV, CO_2 , and $PM_{2.5}$ concentrations were much higher than the Chinese standard levels, specifically in the case of natural ventilation (3300 ppm) (Sui et al., 2021). Moreover, in a similar study, Deng et al. (Deng et al., 2021) conducted a comprehensive analysis of IAQ in 30 houses and results showed that the average concentration of formaldehyde (CH_2O) in summer, $PM_{2.5}$ content in spring, and benzene (C_6H_6) concentration in winter were $100.76 \mu g/m^3$, $80.4 \mu g/m^3$, and $8.46 \mu g/m^3$, respectively, exceeding the Chinese standards as well as the international health guidelines and standards.

Over the past decades, global awareness of sick building syndrome (SBS) in indoor buildings caused by poor ventilation systems has increased. The United Arab Emirates government adopted its own rules and regulations following the emergence of SBS in newly constructed residential buildings in order to promote green buildings and sustainability (Awad & Jung, 2021). SBS, driven mostly by CH_2O and VOCs that are produced by newly constructed building materials, results in symptoms including eye, nose, and throat (ENT) problems, difficulty in breathing, headache, drowsiness, loss of concentration, and chest pain or tightness (Awad & Jung, 2021). Awad and Jung (Awad & Jung, 2021) discussed IAQ in a newly constructed residential buildings and houses in Dubai and found that fifteen out of twenty houses had significantly higher CO_2 concentrations ranging between 682.4 ppm and 2308.2 ppm (Awad & Jung, 2021). Furthermore, TVOC and CH_2O which are the two main pollutants of renovation materials, have exceeded their standard limits in ten and twelve houses, respectively. This result is presumably due to the fact that the frequency of ventilation, such as window opening, decreases in order to avoid a decrease

in indoor temperature in the winter season, when the outdoor temperature decreases. It is clear that in addition to the occupants' difficulty in perceiving air pollution, the lack of knowledge about the ventilation system and the effects of combustion heating might be additional barrier. IAQ can be enhanced by controlling toxic substances and air emissions, as well as regulating temperatures and minimizing humidity in residential homes (Awad & Jung, 2021; Deng et al., 2021). As the green buildings certified by Leadership and Energy in Environmental Design (LEED) have become more common in the United States (Flynn, 2020; Phillips et al., 2020), it is important to highlight that such building innovations serve to improve IAQ and enhance the quality of life for indoor occupants. Recently, new energy-conservative guidelines for new buildings have been implemented in South Asian countries as well, requiring all new buildings constructed in the region to adopt energy-saving strategies to improve sustainability and abide by green-buildings guidelines (Flynn, 2020; Phillips et al., 2020).

In closure, although natural ventilation is energy-efficient, yet it often results in poor IAQ, especially during colder months. Mechanical ventilation and air conditioning systems, if not properly maintained or optimized, can also contribute to pollutant accumulation. Across all studies, $PM_{2.5}$, VOCs, CO_2 , and formaldehyde emerged as the most concerning indoor pollutants, with serious implications for respiratory and overall health. Based on the findings, several key recommendations can be made to improve indoor air quality (IAQ) in residential settings. First, there is a need to promote the adoption of mechanical or hybrid ventilation systems with controlled airflow, particularly during colder seasons when natural ventilation is limited. IAQ assessments should be encouraged in regular and continuous manner to monitor critical pollutants such as $PM_{2.5}$, VOCs, and CO_2 , ensuring early detection and mitigation of health risks. The use of air purifiers equipped with HEPA and activated carbon filters is recommended to reduce both particulate and gaseous indoor pollutants. Additionally, public awareness campaigns should educate residents on safer cooking practices and minimizing the use of indoor combustion sources, especially in poorly ventilated homes and spaces. Building codes and design standards should also be updated to mandate adequate ventilation, pollutant monitoring, and

IAQ performance targets. Finally, IAQ should be integrated into broader public health strategies, particularly in urban areas facing high pollution levels or where housing density increases the risk of indoor pollutant exposure.

3.3 Indoor Air Quality in Schools

IAQ in commercial buildings reveals an area of significant concern, particularly in schools (Kükrer & Eskin, 2021). As IAQ in schools has always been a serious concern for investigators and researchers, this section of the review paper highlights the risk posed by indoor air pollution to occupants of schools including staff, teachers, and children. If the air quality in the school environment is deteriorated, it will have a dramatic effect on the child's health given the fact that students and children spend a lot of time in schools for a period that spans from 6 to 8 h per day (Gupta et al., 2019). While school children are considered vulnerable to IAQ, other office building workers are also exposed to poor IAQ which demands global attention (Fissore et al., 2023). Children under the age of seven in schools are physically more affected by IAQ than adults because their lungs, brains, heart, and other organs are still in the developmental stages and will be more susceptible to indoor air contaminants (Schweiker et al., 2020). People usually spend about 90 percent of their time indoors, and data shows that spending more than 50 h a week is associated with a significant decrease in worker productivity (Schweiker et al., 2020). Table 8 summarizes some recent studies conducted on IAQ in various commercial buildings.

To address the research gap in this field, Kumar et al. (Kumar et al., 2023) discussed the emerging health implications of IAQ and its effects on heart, eye, brain, maternal, and foetal health. The study described several toxic mechanisms that arise from prolonged exposure time to IAQ, and women and children are more susceptible to poor IAQ (Kumar et al., 2023). Similarly, Torriani et al. (Torriani et al., 2023) also conducted a study that examined 26 classrooms in 10 school buildings in the province of Pisa during the winter season (Torriani et al., 2023). The study tested various parameters and discussed the relationship between various parameters with CO₂ concentration which is a parameter often measured as an air quality indicator. Other factors

to be considered during the construction of a building is the ventilation system in the building (Torriani et al., 2023). Ameen et al. (Ameen et al., 2023) investigated thermal comfort and indoor parameters in three office spaces at Gävle University in Sweden (Ameen et al., 2023). Various parameters including CO₂, temperature, and relative humidity were monitored during the field study which was conducted during the spring season (Ameen et al., 2023). The study concluded that the ventilation rate in the two office rooms is less than required guidelines for the offices (Ameen et al., 2023). Furthermore, to better understand the classroom environment during a lecture, Zhang et al. (2023b) assessed the IAQ in a normal classroom located in Beijing. The study measured different parameters of the classrooms repeatedly after five minutes, including CO₂, VOCs, relative humidity (RH), and temperature (T) (Zhang et al., 2023b). The study concluded that average CO₂ concentrations ranged from 1291–1833 ppm and total volatile organic compounds (TVOC) ranged from 159–1178 ppb, indicating inadequate classroom ventilation systems and poor IAQ (Zhang et al., 2023b). However, the study suggested the following possible reason for controlling the low air supply; that is when the classroom temperature rises, the supply of outdoor air should be increased to regulate air cleanliness and freshness in the classrooms and to provide adequate ventilation to maintain a satisfactory IAQ (Zhang et al., 2023b).

Studies have shown that students' learning levels and cognitive abilities can suffer due to poor IAQ, which can get worse by fomites and furniture emissions (Ameen et al., 2023; Torriani et al., 2023; Warner et al., 2017; Wilkins, 2020; Zhang et al., 2023b). In 2023, Jung and Samanoudy (Jung & Samanoudy, 2023), examined the indoor environment and air quality in a university dormitory. The study has shown that only because of the limited living space in these buildings, IAQ in students' residential buildings is expected to be significantly stimulated by furniture and fomites' emissions (Awbi, 2017). Field measurements were made to determine the levels of TVOCs, relative humidity (RH), CO₂, and HCHO content in three identical renovated studio-type rooms located at different locations and on different floors of a new building (Jung & Samanoudy, 2023). Results have shown that TVOC and HCHO concentrations were high in the newly constructed

Table 8 A summary of recent research studies of IAQ in commercial buildings including schools and universities

Ref	Research location	Quantity of samples	Ventilation System	Season	Pollutants Measured
Torriani et al. (2023)	Italian Province of Pisa	26 classrooms in 10 school buildings	Naturally ventilation	Winter	CO ₂
Ameen et al. (2023)	University of Gävle, Sweden	3 office rooms in a historic building	Fan-controlled supply and exhaust and supply air ventilation with heat recovery (FTX system)	April	Relative humidity Temperature, air flow rate, CO ₂ , and room air velocities
Zhang et al. (2023b)	University of Beijing	A total of 10 measurements were completed for 10 consecutive weeks	Naturally ventilation	Spring	Temperature, relative humidity, CO ₂ , and TVOCs
Jung and Samanoudy (2023)	Ajman University in the United Arab Emirates (UAE)	3 rooms in the female hostel building Each room was measured thrice at 1 week to approximately 4 weeks after transfer	Naturally ventilation	Winter	TVOCs, CO ₂ , relative humidity, and HCHO
Abhijith et al. (2022)	London Borough of Lambeth	3 primary schools	Naturally ventilation	Spring and Autumn	PM ₁₀ , PM _{2.5} , PM ₁ , CO ₂ , temperature, and RH
Cabovská et al. (2022)	Gothenburg, Sweden	45 primary school classrooms	Natural ventilation and balanced mechanical ventilation systems (MV) with constant air volume (CAV) and variable air volume (VAV)	NA	Temperature, relative humidity, CO ₂ , PM ₁₀ , PM _{2.5} , TVOC, HCHO, NO ₂ , O ₃ , and Air change rate (ACR)
Cai et al. (2021)	Beijing, China	33 classrooms in 21 schools (5 kindergartens, 8 primary schools and 8 middle schools)	Natural ventilation and mechanical ventilation (MV)	Winter	PM _{2.5} , CO ₂ , temperature, and relative humidity

building in the initial days but decreased after two months (Awbi, 2017). In addition, the study suggested that improving ventilation system and reducing the number of occupants are crucial to control air pollution in this newly constructed building (Awbi, 2017). Due to the absence of modern facilities in rural areas, rural–urban migration is taking place in a quick manner and in unbalanced form (Abhijith et al., 2022; Awbi, 2017; Song et al., 2018). Many educational facilities are built for convenience along major busy roads in urban areas of the UK (Abhijith et al., 2022). This results in the accumulation of large amounts of hazardous substances around schools due to the emission of pollutants due to traffic congestion from busy roads. For example, Abhijith, Kukadia, and Kumar (Abhijith et al., 2022) examined air pollution, ventilation systems, and IAQ in three schools in London. The study focused on optimizing and monitoring parameters including CO₂, PM, and temperature for about two weeks (Abhijith et al., 2022). The study highlights the importance of hazardous contaminants accumulation around schools due to emissions from congested roads which may cause adverse effects on human health. In the latest research, Cabovská et al. (Cabovská et al., 2022) explored the impact of ventilation strategies and IAQ in Sweden primary schools. Three types of ventilation systems with different heating systems have been used to assess IAQ in forty-five primary school buildings (Cabovská et al., 2022). The study found that classrooms with balanced mechanical ventilation systems (MVS) had lower CO₂, HCHO, PM₁₀, and PM_{2.5} levels than classrooms with natural ventilation (NV). They concluded that the adequate ventilation rates in the classrooms can help in reducing contaminant levels. Ventilation strategies reveal varying IAQ levels, emphasizing the importance of balanced mechanical ventilation systems (Cabovská et al., 2022). According to various studies, children may suffer from various diseases due to long-term inhalation of indoor pollutants which may result in reduced cognitive performance and poor learning abilities (Abhijith et al., 2022; Cabovská et al., 2022). The Chinese government has also made great efforts to improve the environments inside educational facilities, particularly in primary schools. In 2021, Cai et al. (Cai et al., 2021) examined various measurements of contaminants including CO₂ and PM_{2.5} along with other parameters such as humidity and temperature in 21 schools in China (Cai

et al., 2021). According to this study, daily measurements of PM_{2.5} concentrations should be <37.5 µg/m³ for 95% of the academic year (Cai et al., 2021). However, none of the classes met the time requirement ratio of ≤37.5 µg/m³. It is important to mention that ventilation systems become a focal point, with chemical emissions and poor ventilation identified as contributing factors for poor IAQ. The study emphasized the need for adherence to governmental standards for CO₂ concentration and daily PM_{2.5} concentrations (Cai et al., 2021). In India, PM emerged as a critical pollutant in schools, with concentrations surpassing national standards and in Delhi, non-commercial buildings faced the brunt of indoor air pollution, with occupant density emerging as a significant factor (Datta et al., 2017). Numerous research studies have shown several benefits of green building regulations (Phillips et al., 2020). The impact of green renovations in New Jersey schools showed a positive outcome, increasing the environmental quality of school buildings and impacting student performance and productivity (Flynn, 2020). However, the research study also highlighted the necessity for more refined assessment techniques to comprehensively evaluate IAQ improvements during the operational phase (Flynn, 2020). It is important to restructure our education system on a flexible and sustainable basis. Therefore, there is an urgent need to develop monitoring programs focusing on measuring hazardous air pollutants and sampling microbial contaminants. This will help in collecting data from various research studies on IAQ in educational institutions such as schools and universities to further improve our knowledge in this diverse field in order to come up with air pollution control programs that can enhance IAQ in most critical indoor environments and promote sustainability in most vulnerable segments of the community.

In conclusion, the reviewed studies provide strong evidence that IAQ in schools is significantly influenced by the type of ventilation system, seasonal variations, and the building's use and occupancy patterns. Naturally ventilated classrooms were consistently associated with elevated levels of carbon dioxide, particulate matter, and other pollutants such as volatile organic compounds and formaldehyde, often failing to meet recommended air quality standards. Mechanical ventilation systems, particularly those with controlled air volume, were found to be more effective in

maintaining healthier indoor environments. Seasonal variations, especially during winter, further exacerbated IAQ issues due to limited air exchange, emphasizing the need for improved ventilation strategies in school settings.

To address the IAQ challenges identified across school environments, several practical and policy-oriented recommendations can be drawn from the reviewed literature. First, it is essential to upgrade school ventilation systems from passive natural methods to balanced mechanical systems. In addition, real-time CO₂ monitoring devices should be installed in classrooms to ensure levels remain below the commonly recommended 1000 ppm threshold. To mitigate exposure to particulate matter, schools should invest in regular maintenance of HVAC systems and consider the use of HEPA filters. During colder seasons, schools should adopt operational guidelines that ensure sufficient ventilation, even with closed windows, by implementing scheduled mechanical airflow or hybrid systems. Furthermore, it is crucial to train school staff and facility managers on IAQ best practices, including how to recognize symptoms of poor air quality and respond appropriately. Lastly, national and local educational authorities should integrate IAQ standards into school health and safety regulations. This includes setting pollutant thresholds, requiring routine IAQ assessments, and allocating funding for ventilation upgrades. By implementing these evidence-based interventions, schools can create healthier, more supportive learning environments for students and staff alike.

3.4 Management Framework

Critical indoor environments such as commercial buildings including healthcare facilities, schools and universities and residential buildings, all require a special and careful attention for ensuring a healthy air quality to protect indoor occupants and users including patients, healthcare workers (HCWs), teachers, staff, elderly people and students like children against chemical pollutants, hospital acquired infections (HAIs) and occupational diseases. Although infection control measures exist and are considered they are often inadequate because of concomitant construction problems and ingress of outdoor pollutants (Baudet et al., 2021; Brambilla & Capolongo, 2019; Chamseddine, 2019; Chamseddine et al., 2019; Facciola et al.,

2019; Gola et al., 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Ibrahim et al., 2022; Kenarkoochi et al., 2020; Odone et al., 2019; Riveron et al., 2023; Taushiba et al., 2023; Uścinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a). This is particularly true in metropolitan congested cities where buildings located in urban environments are surrounded by industrial pollution and vehicular air emissions (Parmaksız et al., 2021). Reduced performance and increased symptoms of sick building syndrome (SBS) such as headache and fatigue have been widely associated with poor IAQ (Arikan et al., 2018; Baudet et al., 2021; Brambilla & Capolongo, 2019; Chamseddine, 2019; Chamseddine et al., 2019; Facciola et al., 2019; Gola et al., 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Ibrahim et al., 2022; Kenarkoochi et al., 2020; Odone et al., 2019; Pathak et al., 2020; Riveron et al., 2023; Taushiba et al., 2023; Uścinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a). In this context, air quality management requires the consideration of a complex array of economic, technical, legal, infection control and political factors to ensure a healthy ambient and indoor environment (Fig. 3). This management framework will be discussed in the following text.

Critical indoor environments such as hospitals and schools are naturally more sensitive to IAQ given the vulnerability of occupants such as patients, elderly people and children. Therefore, the best approach to improve IAQ in critical indoor environments is through three main steps: first is source control, b) dilution with outdoor air via mechanical ventilation, and c) removal of contaminants. Since outdoor air is mostly polluted, thus limiting and removing particles entering from the outdoor environment while also reducing indoor PM sources are crucial and necessary in enhancing the IAQ inside residential buildings as well as commercial buildings like office spaces, sports centers, schools, educational institutions, hospitals and healthcare facilities (Baudet et al., 2021; Brambilla & Capolongo, 2019; Chamseddine, 2019; Chamseddine et al., 2019; Facciola et al., 2019; Gola et al., 2019; Hassan & Zeeshan, 2022; Hiwar et al., 2021; Ibrahim et al., 2022; Kenarkoochi et al., 2020; Odone et al., 2019; Riveron et al., 2023; Taushiba et al., 2023; Uścinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a). Properly designed and maintained central air conditioning and mechanical ventilation systems with adequate

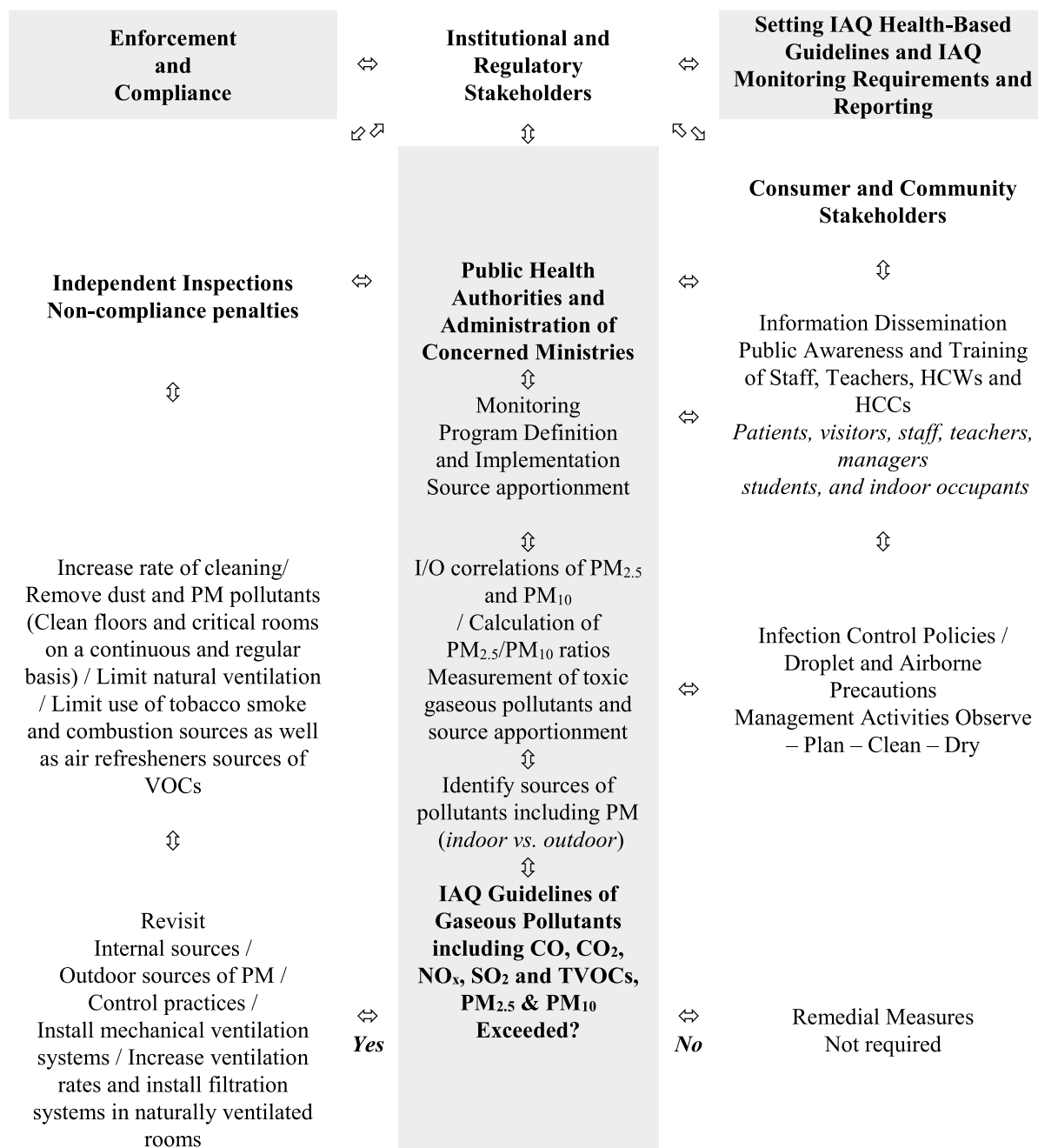


Fig. 3 Air quality management framework

filtration are effective for this purpose (Chamseddine, 2019; Chamseddine et al., 2019; Ibrahim et al., 2022). While natural ventilation can improve air exchange rates, in polluted urban areas it has to be minimized to limit the exposure of patients, employees, and visitors alike to outdoor air pollution levels and traffic air

emissions that often have significant levels of PM₁₀ and PM_{2.5}.

In order to regulate and control air quality inside critical indoor environments such as hospitals it is crucial to consider that hospital-acquired infection (HAI) represent a major public health concern,

where pathogens in healthcare environments can be transmitted through airborne particles, fomites, respiratory droplets or direct contact with bodily fluids (Baudet et al., 2021; Brambilla & Capolongo, 2019; Chamseddine, 2019; Chamseddine et al., 2019; Facciola et al., 2019; Gola et al., 2019; Hassan & Zee-shan, 2022; Hiwar et al., 2021; Ibrahim et al., 2022; Kenarkoohi et al., 2020; Odone et al., 2019; Riv-eron et al., 2023; Taushiba et al., 2023; Uścinowicz et al., 2023; Vonci et al., 2019; Zhang et al., 2023a). Despite the existence of the international infection control guidelines (Chamseddine, 2019; Chamseddine et al., 2019; Ibrahim et al., 2022), an adequate level of basic infrastructure of such policies is still not available in many developing countries with low-income settings. Therefore, preventing infection in healthcare settings requires serious and consistent application of infection control measures by health-care workers (HCWs) and direct involvement of the infection control team. Controlling the spread and transmission of infections and diseases occurring due to viral shredding or bacterial infections in healthcare settings can prevent considerable mortality, morbidity and healthcare costs. Infections are spread through one or more of three main routes: droplet, airborne and contact transmission. Droplets greater than 5 µm in size may be generated from the respiratory tract during coughing, sneezing or even talking (Chamseddine, 2019; Chamseddine et al., 2020; Ibrahim et al., 2022). If droplets from an infected person come into contact with the mucous membranes (i.e. mouth or nose) or surface of the eye of a recipient, they can transmit infection (Chamseddine, 2019; Chamseddine et al., 2020; Ibrahim et al., 2022). These droplets remain in the air for a short period of time and can travel up to 2 m, so physical closeness is required for transmission. As for airborne transmission, aerosol generating procedures (AGP) are considered to have a greater likelihood of producing aerosols compared to coughing for instance. Aerosols are smaller than the droplets and can remain in the air for longer periods, hence potentially transmit infection by mucous membrane contact or inhalation (Chamseddine, 2019; Chamseddine et al., 2020; Ibrahim et al., 2022). Contact transmission may be direct where infectious agents can be inadvertently passed directly from an infected person to a recipient. Indirect contact transmission takes place when a recipient has contact with a contaminated object, such as furniture or object

that an infected person may have coughed or sneezed on (Chamseddine, 2019; Chamseddine et al., 2020; Ibrahim et al., 2022). It is important to mention that infection is greatest in its early stages. For example, the infectious period for influenza virus is thought to be from 1 day before the onset of symptoms until 3–5 days later and for bacterial infection it can be from 1 day and until 3–5 days as well. Children, elderly and seriously ill people may remain infectious for a longer period, and action should be considered to minimize prolonged shedding of influenza virus by patients with risk factors. Evidence shows that viruses can be Transferred from surfaces such as glass or plastic to hands up to 24 h after contamination takes place; also from materials, surfaces and fomites, influenza and respiratory viruses may be transferred for up to 2 h (Chamseddine, 2019; Chamseddine et al., 2020; Ibrahim et al., 2022).

For that reason, hygiene and environmental cleaning are important and necessary in helping to control infection and spread of influenza and respiratory viruses as well as prohibit any bacterial infection in the most critical indoor environments of hospitals and healthcare settings. Therefore, standard infection prevention control precautions are required from all healthcare workers (HCWs) for the care of all patients and patients' environments, to prevent cross-transmission from recognized and unrecognized sources of infection.

Although in the last decades, the indoor air has attained a growing attention, however the legislation and regulations inside critical indoor environments such as hospitals and healthcare facilities, schools and educational institutions as well as residential homes and office spaces are still insufficient. The air quality management framework that is presented here in this review study aims primarily to provide some guidelines for cleaning critical environments such as hospital, schools and residential settings and offer a practical approach to alleviate the risk of nosocomial diseases or HAIs in hospitals and to reduce the risk from environmental contamination and improve IAQ. This management framework must comprise two major activities; cleaning, and inspection and maintenance of ventilation systems.

Frequent cleaning can remove dust and particulate matter, and reduce the bioburden in indoor environments and associated risk of infection and diseases (Chamseddine, 2019; Ibrahim et al., 2022). Cleaning

activities rely heavily upon available resources and managerial support. For example, in hospitals, schools and office spaces, cleaners or janitors receive little or no training for what they do and thus are in need of systematic aid to good practice with in-built risk assessment for themselves, as well as for indoor occupants. The four-steps (Chamseddine, 2019; Ibrahim et al., 2022) of cleaning activities should include the following tasks, the first task in cleaning process consists of a visual or observatory assessment. First of all, inspecting and observing the areas to be cleaned, then considering overall conditions and degree of visual contamination before beginning the cleaning task. For example, in hospital settings, indoor occupants such as patients tend to touch high-risk sites containing dust particles or infection without any hand hygiene reminders. For that reason, cleaners or medical janitors should always evaluate the cleanliness of a room. The overall impression allows a cleaner to initiate cleaning or not, guided by hospital clinical or nursing staff if necessary. Provided cleaning access is timely and appropriate, the decision to clean will be followed by step 2 that requires the cleaner to plan ahead and adjust the area in preparation for the cleaning process. The second task involves cleaning by removing both dirt or dust particles and microorganisms from surfaces, floors and windows, thus reducing the amount of particulate matter and bioburden in indoor environments. Cleaning should always precede disinfection because the presence of dust particles and dirt will hinder disinfectant activity as studies have shown before (Chamseddine, 2019; Chamseddine et al., 2020; Ibrahim et al., 2022; Robertson et al., 2024). Some hospitals and schools use detergents for routine cleaning while others choose products that either inactivate or kill living microorganisms (Chamseddine, 2019; Chamseddine et al., 2020; Fonseca et al., 2022; Ibrahim et al., 2022; Robertson et al., 2024). It is important to mention that detergents should be used only to remove dust and particulate matter while disinfectants are then used for killing microbes. Hence usage of detergents should always precede the disinfectants use. The final task is drying by using clean paper towels or cloths. Drying of cleaning fluids such as detergents and disinfectants on surfaces and floors is also essential since the cleaning process is not complete until all surfaces are completely dry. For example, in hospitals, patient belongings should also be returned to the top of the locker

or bed table, with the crowded site wiped over and similarly allowed to dry. When cleaners or medical janitors leave the patient area, they should apply hand disinfection if the cleaner has to fulfil further duties. In addition, gloves and any other protective apparel or PPE barrier may be removed and hands washed and dried not only in hospital settings but also in office spaces as well as educational institutions and schools. As a final consideration cleaning activities should be done regularly and accurately, and any problems with cleaning should be reported to clinical staff in general and cleaning supervisors in particular.

Ventilation systems in buildings play a major role in providing adequate physical conditions and perceived air quality to indoor occupants and users through fresh air supply, heat removal, and pollutant dilutions (Alazazmeh & Asif, 2021; Awad & Jung, 2021; Awbi, 2017; Chamseddine et al., 2020; Cheek et al., 2021; Flynn, 2020; Jung & Samanoudy, 2023; Kükrer & Eskin, 2021; Phillips et al., 2020). In healthcare and educational facilities, ventilation systems should prevent cross infection risks, harmful emissions, and spreading of pathogens and nosocomial diseases. A good design of the ventilation system can decrease the infection risks as studies have shown (Alazazmeh & Asif, 2021; Awad & Jung, 2021; Awbi, 2017; Chamseddine et al., 2019, 2020; Cheek et al., 2021; Flynn, 2020; Jung & Samanoudy, 2023; Kükrer & Eskin, 2021; Phillips et al., 2020). Thermal comfort variables and physical parameters of critical indoor environments such as hospital and healthcare facilities, schools and educational institutions as well as office spaces are defined by specific ASHRAE standards (Alazazmeh & Asif, 2021; Awad & Jung, 2021; Awbi, 2017; Chamseddine, 2019; Chamseddine et al., 2019, 2020; Cheek et al., 2021; Flynn, 2020; Ibrahim et al., 2022; Jung & Samanoudy, 2023; Kükrer & Eskin, 2021; Phillips et al., 2020). It is important to mention that temperature should be between 21 and 24° C and relative humidity should have a range between 40 and 60% for a better thermal comfort. Air velocities should vary a minimum of 0.05 m/s and can reach up to 0.30 m/s (Alazazmeh & Asif, 2021; Awad & Jung, 2021; Awbi, 2017; Chamseddine, 2019; Chamseddine et al., 2019, 2020; Cheek et al., 2021; Flynn, 2020; Ibrahim et al., 2022; Jung & Samanoudy, 2023; Kükrer & Eskin, 2021; Phillips et al., 2020). to ensure pathogens-free environment. Furthermore, the maintenance team should

change filtration systems on a regular basis to make sure the supply air is free of contaminants once inside indoor spaces. Finally proper and efficient ventilation combined with low-emission building materials can be key factors for an adequate IAQ and control of infections and contaminants spread by air.

In closure, we recommend that public authorities and public health policy makers as well as hospital managers and health-care professionals or practitioners (HCPs) attempt to qualitatively and quantitatively evaluate indoor air of critical indoor environments on a periodic basis, limit natural ventilation, use mechanical ventilation and air purification equipment in during design, maintenance and construction phases of both residential and commercial buildings particularly in hospitals and schools of high vulnerability. Finally embedding such a systematic cleaning protocol in high performance buildings such as hospitals and schools and applying the tasks presented above would practically help in improving IAQ and reducing infections and contamination.

4 Conclusion

Research and investigation of IAQ in educational institutions, healthcare facilities, and residential buildings have demonstrated the impact of IAQ on human health and wellbeing of indoor occupants. Across all three settings, common pollutants such as PM_{10} , $PM_{2.5}$, CO_2 , VOCs, and formaldehyde were consistently identified as major concerns, often exceeding recommended health-based guidelines. In residential buildings, poor ventilation practices, seasonal variations, and everyday activities such as cooking contributed to pollutant accumulation, posing risks to human health. In schools, inadequate ventilation, and high occupancy rates, particularly in naturally ventilated classrooms, led to elevated pollutant levels, which have been shown to negatively affect students' cognitive performance, learning outcomes, and well-being. Similarly, in healthcare facilities, microbial contaminants, VOCs, and fine particulate matter were prevalent, with evidence linking poor IAQ to increased health risks, particularly for vulnerable populations such as patients and staff. Research on IAQ in the future needs to be handled with a novel approach to further gain knowledge and better

understanding of air quality in order to address current gaps. Therefore, more work has to be completed assessing the composition of PM including organic and elemental analysis to identify potential sources of indoor $PM_{2.5}$ and PM_{10} with an emphasis on bioaerosols and biological burden of indoor environments particularly bacteria, fungi and molds. Such studies can be helpful in risk assessments to further develop control measures towards improving IAQ in critical settings such as hospitals and schools and protecting vulnerable individuals such as patients, medical employees, visitors, students, and children alike.

IAQ research brings promising opportunities to further develop our knowledge and design effective strategies and technologies that will lead to healthier and much safer indoor environments. Keeping an eye on the future, several important areas of focus emerge, which call for further elaboration and consideration. A significant area of future research includes the implementation of state-of-the-art ventilation technologies. The study and application of modern ventilation systems, namely intelligent ventilation and air purification technologies can be comprehensive in mitigating and controlling indoor air pollutants and contaminants. The new systems and ventilation technologies are designed to respond to air quality data and occupancy patterns in real-time, and such systems will dynamically achieve desired IAQ conditions and comply with air quality standards to promote thermal comfort and sustainability of buildings. Many of these technologies are in line with the development of smart and green buildings and may bring a real change in the way indoor environmental quality (IEQ) is approached. Adaptation to seasonal variability, changes in building usage patterns, and sustainability of deployed strategies should be measured in more detail and greater efforts over extended time frames. Most of the developed countries evaluate numerous factors and air quality indicators during the planning and construction phase of the building and follow all the needed IAQ regulations and environmental conditions of the indoor space. However, in the case of developing countries, it is somehow clear that the authorities do not consider and implementation of ISO environmental standards and regulations have been found limited and must be developed to further improve IAQ and living standards. With

interdisciplinary collaboration and cooperation, IAQ studies will help in research advancement and would further promote occupational health and safety. The collaboration and coordination of air quality experts and engineers across different sectors like architecture, environmental engineering, health sciences, and policy makers, will have a great potential to lead to smart, comprehensive and integrated solutions.

In closure it is noteworthy to mention that methods to improve IAQ can subsequently be developed through technologies such as information provision, artificial intelligence (AI) and automation systems. It is important to utilize these methods to raise awareness of IAQ and to inform occupants about the harmful effects of pollutants that they face every day. Research into technologies and automation systems can further develop building management systems (BMS) with advanced capabilities, air quality control programs, eco-friendly strategies, energy efficiency, improved building comfort, and low energy costs that serve indoor occupants and their environment in a viable manner. Finally, IAQ real time monitoring in the context of smart building technologies can change the way IAQ is perceived, managed optimized and controlled.

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Data Availability This study is a systematic review research work and doesn't contain any supporting material data. However, all journal articles and papers that were used in this study are available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

Competing interests All authors declare that they have no conflict of interests.

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